Seoul National University
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Falcon9 Simulation

Topics in Aerospace Engineering - Team 5

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1 Introduction

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2 Vehicle, Stages and Mission

The simulator follows the Falcon 9 launch vehicle in the configuration flown on SpaceX CRS-10, the tenth Commercial Resupply Services mission to the International Space Station (launched 19 February 2017 from Launch Complex 39A; the first stage returned to a landing site near the pad) [1]. The model is not a reconstruction of flight hardware: it is a project-configured representation assembled from published sources. Every adopted parameter value therefore carries an evidence class — K (authoritative known value), E (literature estimate) or A (explicit modeling assumption) — and the tables in this report state the class of every value. The underlying records are `data/vehicle_parameters.json` (parameter set and sources) and `data/crs10_mission.json` (mission configuration).

2.1 Vehicle configuration and stages

The modeled vehicle is a two-stage stack of 62.4 m model height: stage 1 (42.6 m), stage 2 (12.6 m), and the Dragon spacecraft with its trunk represented as one uniform assembly of 7.2 m equivalent length and 1.83 m equivalent radius. All axial stations are measured from the structural base of the referenced stage along the vehicle's Body $+x$ axis (toward the nose). Table 1 lists the adopted geometry and mass values.

Component	Value	Class
Outer diameter	3.66 m	K
Stage-1 length	42.6 m	E/A
Stage-2 length	12.6 m	E/A
Dragon+trunk equivalent length	7.2 m	E/A
Model total height	62.4 m	E/A
Stage-1 dry mass	22 200 kg	E
Stage-1 propellant load	410 900 kg	E
Stage-2 dry mass	4 000 kg	E
Stage-2 propellant load	107 500 kg	E
Dragon loaded assembly (incl. cargo)	8 430 kg	E
Launch mass (model sum)	553 030 kg	E/A
Stage-1 dry CG station above base	17.52 m	E/A
Stage-1 dry inertia $[I_{xx}, I_{yy}, I_{zz}]$	[61 368.59, 4 441 994.04, 4 441 994.04] kg m^2	E/A

Table 1 Vehicle geometry and mass model with evidence classes. Stage lengths combine legacy Full Thrust geometry with a contiguous-stage model; the Dragon assembly mass (including 2 490 kg of cargo [2]) is an external estimate [3]. Sources: geometry and dry/propellant masses [5, 7, 4].

Inside stage 1, the RP-1 tank occupies the axial interval $x = 4.0$–18.7 m and the LOX tank $x = 18.7$–40.8 m; the liquids are modeled as settled flat cylinders at constant densities (820 kg/m^3 for RP-1, 1 250 kg/m^3 for LOX). The stage-1 dry mass contains a residual hardware budget of 6 450 kg distributed uniformly over $x = 4.0$–40.8 m, since no surveyed location exists; if that budget were free to move within its interval, the dry CG station would shift between 12.18 m and 22.87 m. This range is a conditional spread of the placement assumption, not a confidence interval. Tank geometry, depletion and the resulting CG and inertia migration are treated in the propulsion and mass model chapter of this report.

2.2 Propulsion inventory

Stage 1 carries nine Merlin 1D SL engines on a 1.3 m radius ring (A); stage 2 carries a single MVac engine. Engine thrust application (pivot) stations are 1.0 m and 0.5 m above the respective stage base (A), and the gimbal limit is 5° (A). The model derives the vacuum thrust from the historical sea-level operating point of 756 kN per engine [6] using the nozzle exit area A_e and sea-level pressure $p_0 = 101\,325$ Pa:

$$T_{\text{vac}} = T_{\text{sl}} + A_e\, p_0, \qquad I_{\text{sp,sl}} = I_{\text{sp,vac}}\, \frac{T_{\text{sl}}}{T_{\text{vac}}} \tag{1}$$

which yields 824.901 kN and 285.023 s (respectively 935 kN and 348 s for the literature-sourced MVac). Table 2 summarizes the engine set; none of these values is a certified CRS-10 engine rating.

Parameter	Merlin 1D SL (×9)	MVac (×1)
Sea-level effective thrust	756.000 kN (E/A)	—
Vacuum thrust	824.901 kN (E/A)	935 kN (E)
Vacuum specific impulse	311 s (E)	348 s (E)
Sea-level specific impulse	285.023 s (E/A)	—
Nozzle exit area	0.68 m^2 (E)	4.90 m^2 (E)
Minimum throttle	60 % (E)	40 % (E)

Table 2 Engine model with evidence classes. The sea-level operating point is a historical SpaceX value [6]; engine data and throttle limits are research-model estimates [7], not certified flight limits.

The initial stage-1 inventory is 123 500 kg of RP-1 and 287 400 kg of LOX [9] (loaded O/F mass ratio 2.327), while the consumption model uses a constant O/F mass ratio of 2.35 [8]. The adopted pair depletes LOX before RP-1 and leaves 1 202.13 kg of RP-1 at the mathematical burn limit. This residual is a consequence of the adopted inventory/ratio combination, not a known Falcon 9 landing reserve; independent tank states would be required for variable O/F, venting or refilling.

2.3 Return propellant budget

The first-stage return reserves propellant by planning allocation (all class A): 38 t for the boostback burn, 16 t for the entry burn, 10 t for the landing burn, and a protected reserve of 2 t. Unused boostback and entry allocations transfer to the landing burn; the reserve and the limiting-tank residual are never consumed. These are planning budgets, not measured characteristics of the CRS-10 vehicle.

2.4 Mission phases and flight paths

The mission executes as an event-driven phase sequence: each transition carries a condition (a fixed reference time, or a guard signal such as altitude, speed, propellant state or landing prediction) and is checked against the accepted integration state. The ascent commands MECO at 140 s (with a 1 s shutdown ramp) and separation at 144 s; these are approximate published preflight times adopted as commands [1]. All later event times are simulated outcomes of the model, not observed CRS-10 event times. Table 3 lists the nominal phases of the full mission.

Three flight paths leave the pad together and diverge at separation (Figure 1). The full stack ascends to MECO and separates; the separated upper stage (stage 2 with the Dragon assembly) ignites 8 s after separation and, under its assumed command sequence, shuts down at SECO 545 s at about 200 km altitude and 1 600 km surface distance from the launch site. The booster flies the return arc: after boostback it coasts to an apogee of about 111.5 km at roughly 90 km surface distance, descends through the entry burn and the aerodynamic descent, and lands at the assumed ground target 14.0 km south and 6.0 km east of the launch site (the nominal run reaches that target within meters). The upper-stage commands are demonstration assumptions (A), not reconstructed flight data; quantitative comparison of the ascent against the video-derived telemetry record and the landing acceptance metrics are discussed in the results chapter of this report.

Phase	Transition condition	Nominal time [s]
Powered ascent	Liftoff initial state	0
Shutdown ramp / MECO	Fixed command, 1 s ramp	140.00–141.00
Full-stack coast	Zero thrust and propellant flow	141.00–144.00
Stage separation	Zero flow, engines off	144.00
Boostback burn	Ignition after fuel and attitude checks	161.00–208.81
Return coast	—	208.81–370.14
Entry burn	Descending through 45 km; ends at 500 m/s	370.14–392.95
Aerodynamic descent	—	392.95–426.16
Landing burn	Predicted stopping clearance + 1.5 km	426.16–465.08
Touchdown	Lowest leg tip reaches the ground sphere	465.083

Table 3 Mission phases of the full simulation case. Times after boostback ignition are simulated outcomes (nominal run, seed 1); the ascent times are adopted reference commands.

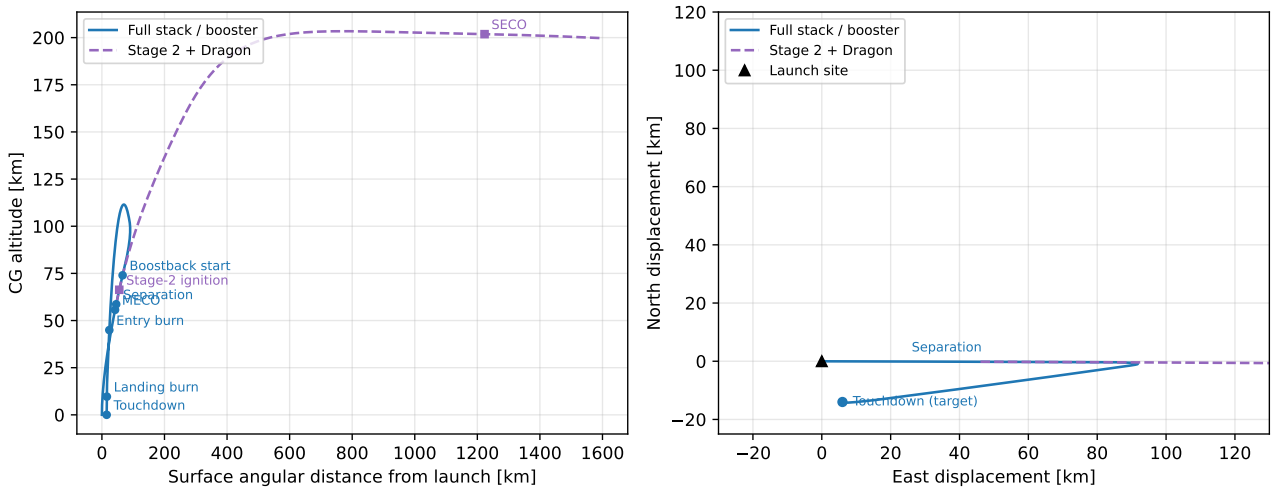


Figure 1 Simulated flight paths of the nominal full-mission run (seed 1). Left: CG altitude against surface angular distance from the launch site. Right: ground displacement; the launch site is at the origin and the assumed landing target lies 14.0 km south and 6.0 km east of it. Markers denote mission events.

3 Environment Models

The environment module (`Environement.py`) is a pure service layer: it supplies the world the vehicle flies through and never calls back into vehicle, dynamics or control code. Its services are the standard atmosphere, point-mass gravity with its gradient, the acceleration of the rotating Earth-fixed frame, the wind, and the ground reference; all evaluations are stateless functions of the current state. Following the project convention, any state outside a modeled domain is rejected with an error, never extrapolated.

3.1 Atmosphere

The atmosphere follows the US Standard Atmosphere 1976 [10]: seven gradient layers between sea level and the mesopause at 84.852 km, reproduced layer by layer. Within a layer with base temperature T_b, base pressure p_b and lapse rate L, and in the isothermal case, the pressure is

$$p(h) = p_b \left(\frac{T_b + L\,(h - h_b)}{T_b} \right)^{-\frac{g_0}{R L}} \quad (L \neq 0), \qquad p(h) = p_b \, \exp\left(-\frac{g_0\,(h - h_b)}{R\,T_b} \right) \quad (L = 0), \tag{2}$$

with $g_0 = 9.80665\,\mathrm{m/s^2}$ and the gas constant of dry air $R = 287.05287\,\mathrm{J/(kg\,K)}$. Density and speed of sound follow the ideal-gas and isentropic relations, $\rho = p/(R\,T)$ and $a = \sqrt{\gamma R T}$ with $\gamma = 1.4$. Sea-level values are $T = 288.15\,\mathrm{K}$, $p = 101\,325\,\mathrm{Pa}$, $\rho = 1.225\,\mathrm{kg/m^3}$.

Above the mesopause the model deliberately stops claiming physics: an isothermal tail at $T_m = 186.946\,\mathrm{K}$ decays exponentially with a scale height of $R\,T_m/g_0 \approx 5.47\,\mathrm{km}$ toward numerical vacuum (density below $10^{-150}\,\mathrm{kg/m^3}$ at the 2 000 km envelope ceiling; $p(200\,\mathrm{km}) \approx 2.7 \times 10^{-10}\,\mathrm{Pa}$). The envelope 0–2 000 km is the mission scope; altitudes outside raise a domain error instead of a number. Figure 2 shows the model against the published USSA 1976 checkpoint values that the test suite pins (5–80 km, altitudes, temperatures, pressures and densities compared at 5×10^{-5} relative tolerance). The environment suite of 41 tests additionally checks layer-boundary continuity, monotone pressure and density decay, exact tail scale height, and hydrostatic balance $\partial p/\partial h = -g_0\,\rho$ inside every layer.

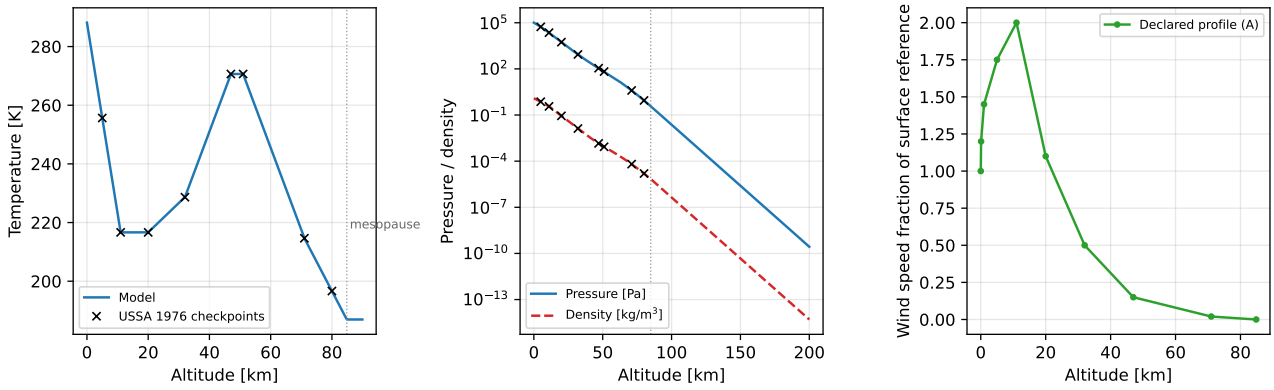


Figure 2 Environment model. Left: layer temperatures to the mesopause with published USSA 1976 checkpoint values. Center: pressure and density (logarithmic) to 200 km, showing the isothermal tail above the mesopause; black markers are the published checkpoints. Right: the declared wind-profile scale factor of the surface reference wind (class A), which is zero above the modeled atmosphere; the nominal mission runs in calm air.

3.2 Gravity and rotating-frame accelerations

Gravity is the reduced Newtonian point-mass model. With the Earth radius $R_E = 6\,371\,\mathrm{km}$ and the configured surface gravity g_0, the gravitational parameter is $\mu = g_0 R_E^2$, so the gravitational acceleration at the launch site is the configured value by construction:

$$\mathbf{a}_g = -\frac{\mu}{\|\mathbf{r}\|^3}\,\mathbf{r}. \tag{3}$$

Gravity never appears as a force; Equation (3) is added once as an acceleration next to thrust over mass. The module also provides the 3×3 gravity-gradient map $\nabla_{\mathbf{r}} \mathbf{a}_g = (\mu/\|\mathbf{r}\|^3)\,(3\,\hat{\mathbf{r}}\hat{\mathbf{r}}^{\mathsf{T}} - \mathbf{I})$, used only in the navigation filter's prediction Jacobian (the navigation chapter), not for torques. Oblateness (J_2) is out of scope.

The external frame is Earth-centred and launch-aligned (detailed in the reference-frames chapter of this report); it does not rotate for the flight mechanics themselves. Earth rotation enters translation only, through the separate frame acceleration

$$\mathbf{a}_{\text{fr}} = -2\,\boldsymbol{\omega}_E \times \mathbf{v} \;-\; \boldsymbol{\omega}_E \times (\boldsymbol{\omega}_E \times \mathbf{r}), \tag{4}$$

the Coriolis and centrifugal terms for velocity relative to the rotating Earth, with the rotation vector $\boldsymbol{\omega}_E = \Omega\,(\cos\varphi,\ 0,\ \sin\varphi)$ at launch latitude φ. The nominal mission uses the sidereal rate $\Omega = 7.2921150 \times 10^{-5}$ rad/s at $\varphi = 28.6°$; setting $\Omega = 0$ disables the terms. The frame accelerations, and their analytical gradients, are supplied once per integration step next to gravity. The angular equations and the gyro model retain the project's approximate Earth-rate treatment: no complete inertial-rate coupling is claimed.

3.3 Wind and ground reference

The wind model is a surface reference vector (ECEF components in the mission configuration) that can be rotated about the launch-site vertical, scaled by a declared altitude profile: the scale factor of the surface reference grows through the boundary layer, peaks at 2.0 around the 11 km jet-stream core, decays through the stratosphere and reaches exactly zero at the mesopause — there is no wind above the modeled atmosphere (Figure 2, right). The profile is a declared stylization (class A). There are no gusts, no time dependence and no direction shear, and the wind enters the physics only through the relative-velocity term of the aerodynamics model. The nominal full-mission run uses calm air: the mission configuration sets no surface reference wind, so the default zero vector applies throughout.

The ground is a bare sphere of radius R_E; altitude is measured radially above it, with an optional reference-level shift, and supports a station offset from the center of gravity (used, for example, for engine-exit altitude). Terrain and contact forces are out of scope; contact is evaluated at the mission level through the leg-tip geometry discussed with the landing criteria in the return-control chapter of this report.

The module validates its configuration at construction: non-finite gravity, radius, ground offset or wind, non-positive radius, a negative rotation rate or a latitude beyond ±90° all raise an error before flight, not mid-integration.

4 Aerodynamics

The aerodynamic model (`Aerodynamics.py`, data in `data/aero_configurations.json`) supplies aerodynamic forces and moments in current Body components, Body $+x$ toward the nose, stations measured from the structural datum of the active configuration. One model instance serves the whole flight through five declared vehicle configurations, switched by the mission layer at phase transitions; every coefficient entry in the data file carries an evidence and uncertainty record (the evidence-class convention of Section 2).

4.1 Flow state and force build-up

All aerodynamic terms act on the relative flow velocity

$$\mathbf{v}_{\text{rel}} = \mathbf{v}_{\text{CG}} - \dot{\mathbf{c}} - \mathbf{w}, \qquad \alpha = \operatorname{atan2}\big(\lfloor \mathbf{v}_{\text{rel},yz} \rfloor,\ v_{\text{rel},x}\big), \tag{5}$$

where $\dot{\mathbf{c}}$ is the axial center-of-gravity migration rate (Section 2, stage depletion) and $\mathbf{w}$ the wind in Body components (Section 3). The total flow angle α spans 0 (nose-first) to π (engine-first); every direction in the lateral plane is handled, not only a pitch plane. The dynamic pressure is $\bar{q} = \frac{1}{2}\,\rho\, v_{\text{rel}}^2$ with the atmosphere state of Section 3.

The model decomposes the load into an axial, a potential-normal and per-segment crossflow terms:

$$F_x = -\,\bar{q}\,S_{\text{ref}}\,CA_0(M)\ \cos\alpha\,|\cos\alpha|, \qquad N = \bar{q}\,S_{\text{ref}}\,C_{N\alpha}\ \sin\alpha\,|\cos\alpha|, \tag{6}$$

$$D_i = \bar{q}\, C_{Dc,i}\, S_i\, \sin^2\alpha \qquad \text{per side segment } i, \tag{7}$$

with reference area and length per configuration (Table 4). The normal force acts along the negative lateral relative-velocity direction at the configuration's windward-end station (57.6 m full stack, 41.0 m booster), mirrored about the center-of-gravity station for retro flow; crossflow segments act at their side-area centroids. For `booster_landing` the body segment (width 3.66 m, $C_{Dc} = 1.1$) is supplemented by the combined grid-fin segment (38.6–40.6 m, 2.4 m, $C_{Dc} = 1.0$) and the deployed-leg segment (0–3.6 m, 1.6 m, $C_{Dc} = 1.2$).

The $\cos\alpha\,|\cos\alpha|$ factor appears in Equation (6) on purpose: the axial force scales with the axial dynamic-pressure share and reverses direction for retro flight, and the grid-fin effectiveness below scales the same way, so a fixed deflection command reverses its steering effect in engine-first flight and vanishes at $\alpha = 90°$. Station moments are computed once as (station − CG) × **F** and returned as a force through the center of gravity plus a free moment, so the load interface cannot double-count. Windward-end pressure centers toward the nose are destabilizing in nose-first flight; deployed hardware near the trailing end stabilizes tail-first descent. The model reproduces this textbook behavior from the stated geometry — it does not stabilize the vehicle on its own.

Rotary damping closes the moment set:

$$\mathbf{M}_{\text{damp}} = c\,\frac{\bar{q}\, S_{\text{ref}}\, L_{\text{ref}}^2}{2\, v_{\text{rel}}}\,\boldsymbol{\omega}, \tag{8}$$

with pitch and yaw tied by symmetry and roll treated separately (declared coefficients c: −10 full stack, −20 booster, roll −3 with fins deployed).

4.2 Configurations and axial curves

Configuration	Phase	S_{ref} [m^2]	L_{ref} [m]	Axial curve / fins
`full_stack`	launch to separation	21.08	62.4	sourced M 0–7; declared M 7–10
`booster_alone`	coast, boostback	10.52	42.6	declared (±50 %)
`booster_return`	entry burn, descent	10.52	42.6	declared; four fins
`booster_landing`	landing burn	10.52	42.6	declared; four fins and legs
`second_stack`	stage-2 body (declared)	10.52	19.8	declared, sleeker estimate

Table 4 The five aerodynamic configurations. Station datum is the configuration's own structural base. Fins deploy after boostback; the legs deploy at landing ignition, which is represented by the segment difference between `booster_return` and `booster_landing`.

Figure 3 shows the adopted axial coefficient curves $CA_0(M)$. Only the full-stack curve is sourced: a raster readout of Pezent's RasAeroII analysis (0° angle of attack, full Falcon-9-like stack with fairing) [7], whose transonic peak $CA_0 \approx 1.16$ near $M \approx 1.06$ carries the worst readout uncertainty. Beyond Mach 7 the curve continues as a declared slow hypersonic decay to $CA_0 = 0.75$ at Mach 10, where the vehicle flies above roughly 55 km and axial drag stays below about 0.5 % of thrust. All other curves are declared engineering estimates: a blunt-ended cylinder for the booster ($CA_0 = 0.42$ subsonic, raised to 0.67 by the flat fin/leg increment), a sleeker estimate for stage 2 plus Dragon (0.32 subsonic), uncertain to roughly ±50 %. They close the simulation loop and produce plausible dynamics; they are not good enough to predict quantitative trajectories beyond the sourced regime. Replacing them is a data change in the JSON, not a code change.

4.3 Grid fin control

The `booster_return` and `booster_landing` configurations declare four grid fins at intercardinal azimuths (45°, 135°, 225°, 315° about Body +x), mounted at station 39.6 m on a 1.83 m pivot radius; each fin is modeled as a porous lattice panel of 1.65 m^2 with lift slope $c_{\ell,\delta} = 1.2$/rad (about half of a solid panel of aspect ratio

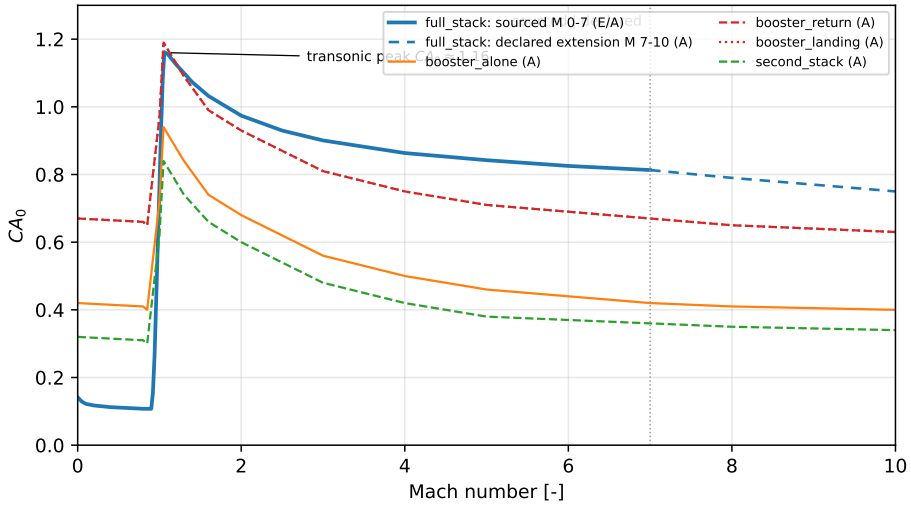


Figure 3 Axial drag coefficient $CA_0(M)$ of the five configurations. Only the full-stack curve to Mach 7 is sourced; everything else is a declared estimate (evidence classes in the legend). The transonic peak of the full stack reaches $CA_0 \approx 1.16$.

near one, because of lattice porosity) and a travel limit of ±0.35 rad (≈ ±20°). A positive deflection of fin i produces the force

$$\mathbf{F}_i = \bar{q}\, S_{\text{fin}}\, c_{\ell,\delta}\ \delta_i \cos\alpha\, |\cos\alpha|\ \big(0, -\sin\varphi_i, \cos\varphi_i\big), \tag{9}$$

applied at the fin's three-dimensional mount station, so the axial lever carries pitch/yaw and the radial offset carries roll — the moment combinatorics are automatic: equal deflections give pure roll, opposite differential pairs give pure pitch, a single fin gives both. At the declared values, full deflection at Mach 3 and 15 km commands about 1.4 MN m of pitch and 385 kN m of roll, an order of magnitude less above 30 km — consistent with grid fins mattering mainly below entry-burn altitudes. Command shaping (rates, travel dynamics, deadbands) belongs to the actuator layer of the control chain and is not part of this model: the aerodynamics accept instantaneous validated commands only. Mach- and angle-of-attack-dependent fin effectiveness, interference with the passive crossflow segments, lattice state and the deflection axial-drag increment are declared unmodeled.

4.4 Domain and verification

Following the project convention, the model rejects rather than extrapolates: Mach outside the adopted curve (all configurations end at $M = 10$), unknown configuration names, command shape or travel violations, and construction-time station violations all raise errors. Within the domain, declared-estimate coefficients do not fail — they produce plausible numbers with real uncertainty.

The test suite (`tests/test_aerodynamics.py`, 42 tests) pins the conventions class by class: zero-angle flight reproduces the simple drag baseline byte for byte; normal-force direction, retro-flow reversal and the windward-end mirror convention hold in both flow directions; the damping formula is checked term by term; physical invariants (the flow extracts energy at every angle, damping dissipates, wind enters only through the relative velocity, continuity at $\alpha = 90°$) are enforced; the fin combinatorics of Equation (9) are verified at all azimuths including retro reversal; and all five configurations load from the JSON with the sourced knot intact, while a two-second vertical propagation exercises the provider integration into the shared dynamics core.

5 Ascent, Staging and Upper Stage

This chapter describes the command laws behind the mission phases of Section 2: the ascent attitude program, the engine shutdown, the separation state map with its audit, and the separated upper-stage continuation. The

event machine, step control and state bookkeeping of the run loop are described in the architecture chapter of this report.

5.1 Attitude program and ascent control

Powered ascent flies a pitch program with an ideal attitude-control law. The commanded tilt grows from vertical along

$$\theta(t) = \theta_{\max} \frac{x^2}{1 + x^2}, \qquad x = \frac{t - t_s}{\tau}, \tag{10}$$

with $t_s = 8\,\mathrm{s}$, $\tau = 100\,\mathrm{s}$ and $\theta_{\max} = 80^\circ$ toward a 45° north-east heading (an International Space Station (ISS)-launch approximation). The tilt law is calibrated against the video-derived ascent reference used in the results chapter; it is not independently validated guidance (class A). Attitude tracking applies a proportional-derivative law on the Body-frame attitude error $\mathbf{e}$ and angular rate,

$$\boldsymbol{\alpha} = k_p\,\mathbf{e} - k_d\,\boldsymbol{\omega}, \qquad \mathbf{M} = \mathbf{J}\,\boldsymbol{\alpha}, \quad k_p = 2.0,\ k_d = 2.5, \tag{11}$$

an unlimited ideal moment: the ascent controller relies on the actuation chain only for realistic execution, and the provider optionally enforces a declared angle-of-attack limit, raising a domain error when the flight exceeds it.

5.2 Engine shutdown and event guards

Engine cutoff at 140 s follows a quintic flow envelope, not an instantaneous stop:

$$\gamma(u) = (1 - u)^3(1 + 3u + 6u^2), \qquad \gamma'(u) = -\frac{30\,u^2(1 - u)^2}{T}, \quad u = \frac{t - t_0}{T}, \tag{12}$$

with $T = 1\,\mathrm{s}$ and completion anchored to the published 141 s MECO reference. The envelope and its derivative are supplied together to the integrator, keeping mass flow and flow rate consistent during the ramp; the duration is a numerical design choice of class A, anchored to, not measured from, the reference timeline. Event transitions carry guards that must hold at the event: separation requires zero flow, engines off and the full stack attached; boostback, entry and landing ignitions additionally require the respective propellant budget (including the reserve), attitude readiness and propulsion readiness, and for the latter two a descending trajectory. Each burn ends through a smooth ramp with its own reason: predicted-miss tolerance, speed target, or budget cap.

5.3 Stage separation and conservation audit

Separation at 144 s is a zero-impulse, no-discard state map: the coasting parent is split into booster and upper stack at the stage-2 interface, both children keep the parent attitude and angular rate, and the upper stage's stations are renumbered to its own aft-skirt datum. The map refuses to run while any engine command or flow is active, and its result is only returned if a conservation audit confirms continuity to tight tolerances: mass, linear momentum, angular momentum about the Earth-centred origin and kinetic energy are conserved to 10^{-10} relative residual, fixed structural points keep their ECEF position and velocity to $10^{-8}\,\mathrm{m/m/s}$, and attitude and angular rate are preserved to 10^{-12}. The audit quantities (all residuals, and both child masses) are recorded in the run's event record, and the results chapter quotes them.

5.4 Upper-stage continuation

After separation the upper stack continues under its own reduced provider as a six-degree-of-freedom propagation from the conserved separation state: a single MVac engine at 0.95 throttle with quintic ignition and cutoff ramps (the standard $10u^3 - 15u^4 + 6u^5$ envelope) over 1 s at 152 s ignition and 545 s cutoff, an attitude law of the form of Equation (11) ($k_p = 2.0$, $k_d = 2.5$) holding the separation heading, and a simple axial-drag

estimate of $C_D = 0.3$ on the 10.52 m^2 stage cross-section — deliberately not the declared `second_stack` aerodynamic configuration of Section 4. These are demonstration commands of class A, not reconstructed flight data: the continuation exists to show the separated state and the conserved separation map, and its SECO state (about 200 km altitude at 1 600 km surface distance in the nominal run) feeds the flight-path overview of Figure 1 rather than any orbital-accuracy claim.

6 Booster Return, Control and Actuation

The return controller commands the separated booster from separation to touchdown through the phases of Section 2: boostback burn, return coast, entry burn, aerodynamic descent and landing burn (`recovery.py`). Control is recomputed on a fixed guidance period of 0.2 s; throttle changes run through 1 s smooth ramps, and the burn timing of the nominal run appears in Figure 4. Boostback and entry burns run the three-engine set $(0, 1, 5)$; the landing burn runs the single center engine. The legs deploy at landing ignition. The controller feeds back the truth state and commands ideal attitude moments — class A project assumptions — so the navigation solution is deliberately not in the loop yet.

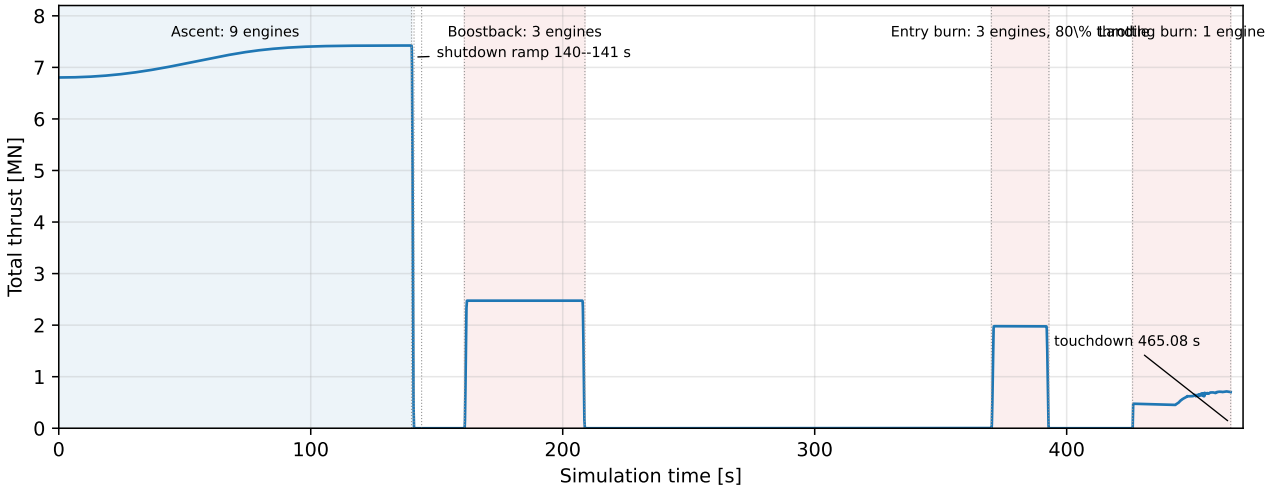


Figure 4 Total thrust of the nominal full-mission run (seed 1). The booster burns are command-planned: boostback on three engines at full throttle, the entry burn on three engines at 80 % throttle, and the landing burn on the single center engine. Touchdown at $t = 465.08$ s.

6.1 Boostback: prediction and steering

Boostback steering inverts a predicted landing miss. A reduced-order ballistic predictor propagates the booster state forward in 0.5 s steps with central gravity and rotating-frame accelerations (Section 3), the declared `booster_return` axial and crossflow coefficients (Section 4), and a prospective three-engine entry burn at the declared altitude and throttle; the resulting predicted impact miss is refreshed every second. During coast and the burn, the commanded acceleration is

$$\mathbf{a}_{\mathrm{cmd}} = -\frac{\hat{\mathbf{m}}}{8\,t_f} + 0.8\,\hat{\mathbf{u}}, \tag{13}$$

where $\hat{\mathbf{m}}$ is the predicted miss vector, t_f the ballistic flight time and $\hat{\mathbf{u}}$ the local vertical. The burn stops on the first satisfied condition: the predicted miss norm falls below 500 m (the "predicted ballistic miss tolerance"), or the burn duration leaves its 10–60 s planning window, or its budget cap is reached. In the nominal run the boostback ends at 208.81 s on the planning cap.

6.2 Entry burn and aerodynamic descent

Entry ignition requires the descending-through-45 km condition plus the restart guards (budget, attitude, propulsion). During the burn the commanded acceleration is a 30 m/s^2 retrograde bias plus lateral correction, $\mathbf{a}_{\text{cmd}} = 30\,\hat{\mathbf{u}} + (\mathbf{v}_{d,h} - \mathbf{v}_h)/8 - \mathbf{a}_{\text{aero}}$, toward the desired horizontal velocity $\mathbf{v}_{d,h} = -\mathbf{e}_h/(t_f + 25\,\text{s})$ that nulls the target error over the remaining flight. The burn ends when the speed reaches 500 m/s or its budget cap fires. Through the aerodynamic descent the vehicle holds a retrograde attitude above 8 km and transitions to vertical attitude below, with the grid fins available for residual moment demand.

6.3 Landing burn and touchdown

Landing ignition is predictive: a reduced-order stopping simulation propagates the booster in 0.1 s steps with the landing aerodynamic model, tilted thrust scaled by the ignition factor 1.05, and the current throttle amplitude; when the simulated stop leaves the vehicle no more than the declared 1.5 km clearance allowance above the ground, the burn ignites (nominal: 426.16 s). The guidance law is a second-order polynomial descent to the target with terminal vertical velocity $v_t = -0.5$ m/s:

$$\mathbf{a}_{\text{cmd}} = \frac{6\,(\mathbf{r}_{\text{goal}} - \mathbf{r})}{t_{go}^2} - \frac{4\,\mathbf{v}}{t_{go}} - \frac{2\,v_t}{t_{go}}\,\hat{\mathbf{u}} - \mathbf{g} - \mathbf{a}_{\text{aero}}, \tag{14}$$

with t_{go} from the current altitude and sink rate. The lateral component of Equation (14) is clamped by the 20° tilt limit, which tapers to a 4° terminal attitude below the 20 m hold height. Touchdown is the ground contact of the lowest leg tip — four assumed tips, 1 m below the base at a 3.5 m radius — and the run loop resolves the contact instant by bisection; the acceptance criteria of Table 5 are evaluated at that instant. The command path from guidance through actuation to the applied forces and moments is completed by the actuator layer.

Landing acceptance metric	Limit	Result (nominal run)
Vertical speed	≤ 2 m/s	0.497 m/s
Horizontal speed	≤ 1 m/s	0.902 m/s
Target error	≤ 10 m	3.306 m
Tilt	$\leq 5°$	0.535°
Angular rate	$\leq 1\,°/$s	0.013 °/s

Table 5 Provisional project acceptance limits for a near-stationary upright contact (class A; not Falcon 9 certified limits) and the nominal run's contact metrics at 465.083 s.

6.4 Force and moment allocation

Each commanded acceleration is converted to a bounded force allocation: a numerical solver chooses the nose direction within the active tilt envelope, the engine count and the throttle such that the resultant of engine thrust (vacuum thrust minus exit-area ambient pressure at the engine station) and the modelled aerodynamic reaction matches the requested force; the corresponding ideal attitude moment drives the vehicle toward that nose direction, with the moment authority capped at 2 MN m during burns and 200 kN m in coast. Grid fins carry residual moment demand beyond the virtual-vector authority, mainly roll. Control gains: attitude $k_p = 1.0$, $k_d = 2.0$; landing lateral $k_p = 0.0576$, $k_d = 0.48$; landing attitude $k_p = 1.5$, $k_d = 6.0$.

7 Sensors and Navigation

The navigation subsystem generates sensor measurements from the truth trajectory and estimates the vehicle state with a nine-state Kalman filter. Sampling occurs only at accepted integration times: no random numbers are drawn during integration trial evaluations, so a fixed seed reproduces a run exactly.

7.1 Sensor models

Sensor	Rate	Station [m]	Error model
IMU	100 Hz	$[38, 0, 0]$	accel $\sigma = 0.01\,\mathrm{m/s^2}$; gyro $\sigma = 10^{-4}\,\mathrm{rad/s}$; nominal biases $[0.05, -0.03, 0.02]\,\mathrm{m/s^2}$, scale factor 0.001, bias random walk 10^{-4}, z-axis gyro bias $10^{-4}\,\mathrm{rad/s}$; misalignment and temperature models supported
GNSS	10 Hz	$[41, 0, 0]$	pos $\sigma = [1.5, 1.5, 2.5]\,\mathrm{m}$ plus Gauss-Markov $\sigma = [0.6, 0.6, 1.0]\,\mathrm{m}$ at $\tau = 300\,\mathrm{s}$; vel $\sigma = 0.1\,\mathrm{m/s}$; latency 0.1 s; outages and dropouts supported
LiDAR	10 Hz	$[0.5, 0, 0]$	slant range 1–1 800 m, $\sigma = 0.05\,\mathrm{m} + 10^{-4}\,R$; boresight body $-x$

Table 6 Sensor models and adopted settings. The noise and bias values are placeholder research assumptions (class A/C): not certified Falcon 9 sensors.

Table 6 summarizes the three models. The inertial measurement unit (IMU) measures specific force and angular rate at its fixed structural station; the measurement model includes the station kinematics (angular-rate and center-of-gravity migration terms) of the dynamics core. The global navigation satellite system (GNSS) receiver provides position and velocity in the external frame, with a Gauss-Markov correlated error beside the white noise, lever-arm correction for its antenna station, outage and dropout support, and a latency model that carries and interpolates the delayed samples. The light detection and ranging (LiDAR) measures the slant range to the spherical ground along its boresight; its update uses the analytic intersection geometry with the planet sphere and rejects invalid pointing geometry rather than extrapolating.

7.2 Nine-state navigation filter

The filter (`kalman.py`) estimates IMU-station position, velocity and accelerometer bias in the external frame, $\mathbf{x} = [\mathbf{p}, \mathbf{v}, \mathbf{b}_a]$, with the strapdown prediction

$$\mathbf{a} = \mathbf{R}\,(\mathbf{f}_{\mathrm{meas}} - \hat{\mathbf{b}}_a) + \mathbf{g}(\mathbf{p}) + \mathbf{a}_{\mathrm{fr}}(\mathbf{p}, \mathbf{v}), \qquad \begin{aligned} \mathbf{p}^+ &= \mathbf{p} + \mathbf{v}\,\Delta t + \tfrac{1}{2}\,\mathbf{a}\,\Delta t^2, \\ \mathbf{v}^+ &= \mathbf{v} + \mathbf{a}\,\Delta t, \end{aligned} \tag{15}$$

where the model gravity and rotating-frame acceleration come from the environment model of Section 3, linearized through the same gravity and frame-gradient maps that the environment module supplies. The attitude rotation $\mathbf{R}$ is the truth attitude: the filter does not estimate attitude or gyro bias — a declared, explicit limitation of the current implementation, not an oversight. The process noise follows from the accelerometer power spectral density $\sigma_a^2/f_{\mathrm{IMU}}$ propagated through the strapdown kinematics, plus a bias random-walk term; the initial covariance is $\mathrm{diag}([2, 2, 2,\ 0.05, 0.05, 0.05,\ 0.1, 0.1, 0.1])$ with a seeded initial position error of $[1, -1, 0.5]$ m.

GNSS updates correct position and velocity with the antenna lever arm and rotate the arm with the angular rate; latency is compensated by extrapolating the measurement with the current acceleration estimate. LiDAR updates are scalar and nonlinear, with the measurement Jacobian taken analytically at the sphere intersection. Both updates use the Joseph-form covariance update, and normalized innovation squares are recorded for monitoring. One filter instance and the same first-stage sensor stations run across separation without a reset to the new true center of gravity — the estimated state is the IMU-station state, and that convention is retained deliberately through the staging event.

The sensor and filter chain is verified end-to-end by the cross-module and mission-level test suites; in the nominal full-mission run the navigation position error holds 0.918 m root mean square (RMS) with truth attitude (quoted and discussed in the results chapter of this report).

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